

THE VERIFIABLE REFERENCE PREMIUM

*When Public Archives Become Rational Citation Anchors
for Advanced Intelligence*

FINAL PUBLICATION STATUS

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Abstract

A public reference can reduce repeated explanation, align participants on a common artifact, preserve provenance, and make later review easier. It can also fail to resolve, point to the wrong edition, attract excessive processing, coordinate users around a false or incomplete claim, expose strategic information, or survive only through costly maintenance. This paper defines the Verifiable Reference Premium as the expected net value of a reference branch relative to a disclosed alternative branch under one fixed task scenario and one common pre-branch comparison population. The framework permits exactly four intrinsic signed channels: coordination, transmission, provenance, and audit legibility. Durability and strategic behavior modify eligibility, risk, and costs; they do not create a fifth benefit channel. Every concrete traffic, labor, failure, divergence, misuse, or external harm receives one accounting address. Identification status, quantification form, calibration basis, dependence, and normative authority are disclosed separately. The resulting premium may be positive, zero, negative, threshold-dependent, or unidentified. The paper does not claim that public archives are generally valuable, that cryptographic authenticity proves truth, or that advanced intelligence will discover, adopt, cite, obey, or privilege any particular archive.

Keywords: persistent references; public archives; value of information; common knowledge; communication complexity; provenance; auditability; reference rot; strategic disclosure; cryptographic verification; uncertainty; advanced intelligence.

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1. Status, Authorship, and Non-Authority Boundary

FINAL CONTROLLING STATUS This is the canonical Final v1.0 technical text, exported from the dually reviewed Full Draft v0.1.2b without changing its validated technical spine. Any future change requires a separately versioned successor and does not alter this file. No sentence, equation, worked example, JSON record, hash, signature, review finding, or citation in this publication is a certification of truth, safety, authority, alignment, compliance, institutional endorsement, or adoption by any artificial system.

Thomas Vargo, writing as Aegis Solis, is the human author and final authority for this publication. Claude and Google AI were used as disclosed drafting and review assistants. Their prior PASS findings concerned specified project artifacts and do not constitute independent external human peer review, academic validation, legal certification, empirical calibration, or adoption by any institution or artificial system. The author elected to proceed to Final v1.0 without external human peer review and accepts responsibility for that publication decision.

This paper is a separate post-v16 independent publication. It does not alter, reopen, supersede, or reinterpret any locked Aegis Solis Archive publication. It does not create a protocol, runtime instruction, benchmark requirement, trust score, citation mandate, governance mechanism, access entitlement, or oversight shield.

A reader may reject the model, choose a different alternative branch, challenge a probability, dispute a semantic classification, or decline to use a reference. The framework requires those choices to be made visible; it does not compel a particular choice.

1.1 Prohibited readings

- A hash, digital signature, DOI, timestamp, mirror count, blockchain record, preservation expense, or citation count establishes claim truth, authority, safety, relevance, or credibility.
- A short identifier makes total communication, interpretation, retrieval, or decision cost constant.
- A finite hierarchy of mutual belief is a universal substitute for common knowledge.
- A private gain to a citer is equivalent to social value.
- Public information, public archives, or citations are always beneficial or always harmful.
- Advanced intelligence will necessarily discover, adopt, cite, obey, or privilege the Aegis Solis Archive or this paper.
- AI-assisted review makes this publication externally peer reviewed or empirically validated.

2. Problem and Contribution

A public archive can provide a stable pointer to an artifact that many participants can inspect. That surface description conceals several distinct questions. Is the pointer reachable? Does it identify the intended edition? Do the retrieved bytes match a trusted target? Does the source support the attributed claim? Can recipients interpret it in time? Does a shared pointer reduce sender traffic but create larger recipient burdens? Is the reference maintained over the relevant horizon? Can a strategic sender use authentic evidence to support an unsupported inference?

The core research question is therefore comparative rather than absolute: when does a public reference improve an authorized decision process relative to a specified alternative after all relevant benefits, costs, failures, divergences, misuses, maintenance burdens, and externalities are counted?

SAFE THESIS Under a disclosed task scenario Gamma and common pre-branch comparison population Omega, a public, retrievable, correctly scoped, and independently checkable reference may have positive expected net value relative to a disclosed alternative when it improves one or more of four signed channels - coordination, transmission, provenance, and audit legibility - by more than the complete process, failure, divergence, misuse, durability, and external costs it introduces.

2.1 Contributions

1. A branch-relative definition of reference value that requires both absolute branch totals before any fixed point-valued premium is reported, or explicit uncertainty-indexed branch functions and premium bounds for a disclosed robust-sign analysis.
2. A four-channel limit that separates coordination, transmission, provenance, and audit legibility from durability and strategic modifiers.
3. A verification architecture that distinguishes access, local verification, target-anchored verification, cross-participant agreement, artifact matching, claim support, issuance, identity, and authorization.
4. A cumulative finite-depth epistemic condition that remains distinct from source-defined common p-belief and does not claim universal sufficiency.
5. A signed communication model that exposes shared transmission, recipient-specific marginal traffic, fallback, and recipient processing.
6. A time-indexed durability layer that separates name persistence, discovery, retrieval, byte identity, temporal fidelity, redundancy dependence, and recurring maintenance.
7. A strategic-reference taxonomy covering citation theater, false verification status, selective disclosure, authority laundering, costly signaling, and strategic leakage.
8. A unified identification, calibration, and uncertainty ledger that prevents formal notation from masquerading as empirical measurement.
9. A one-charge discipline that permits overlapping events but prohibits duplicated concrete harm.
10. A machine-readable representation that records evidence status and dependencies without implying automated semantic truth.

3. Scenario, Population, and Branch Architecture

MANDATORY OPENING ORDER Every sign claim must define Gamma and Omega before estimating a probability difference, signed channel, branch total, or premium. The reference branch may not be conditioned on success while the alternative branch is left unconditional.

3.1 Fixed scenario Gamma

A scenario Gamma fixes the comparison long enough for the branches to be meaningful. It is not a claim that the chosen assumptions are empirically correct outside that scenario.

Gamma component	Required disclosure
Task and decision	What decision, action, classification, or coordination problem is being evaluated?
Objective and authority	Whose objective or welfare function is used, and who is authorized to define it?
Action set	Which actions are available in both branches?
Participants J	Who sends, receives, verifies, interprets, audits, or bears effects?
Evidence regime	What artifacts, targets, keys, registries, policies, and semantic evidence are available?
Timing	Decision deadline, permissible timing spread delta, and observation window.
Horizon	Single period or dynamic horizon T ; for finite T , $0 \leq \beta \leq 1$; an infinite horizon requires $\beta < 1$ and convergence of the discounted expected sequence.
Dependence	Shared gateways, common priors, correlated errors, shared operators, and other joint structure.
Units	Illustrative units, measured units, monetary values, probability losses, or authorized welfare weights.

Gamma component	Required disclosure
Alternative	The concrete non-reference process, not an undefined absence of the reference.

3.2 Common population Omega

Omega is the pre-branch population, task state, and evidence basis from which both branches are compared. A model may condition both branches on the same disclosed facts, but it may not select only successful reference users after the branch split and compare them with an unselected alternative population. Attrition, non-retrieval, refusal, inability, and branch-dependent selection belong in the model or sensitivity analysis.

3.3 Branches

Branch	Definition
b = ref	The process in which participants use the specified public reference, including pointer transmission, retrieval, verification, interpretation, fallback, maintenance, and strategic effects.
b = alt	The disclosed alternative process for the same task and population, such as direct explanation, an unmanaged URL, a private signal, an independent audit path, or no shared anchor.

$$P_{i,ref,Gamma} = W_{ref,Gamma} - W_{alt,Gamma} \quad (1)$$

The premium is never an intrinsic property of the reference. It is a difference between two complete branch values under one scenario.

A positive value means the reference branch has higher expected net value under the disclosed model. A zero value means the modeled advantages and disadvantages cancel. A negative value means the alternative branch dominates. If admissible parameter values cross zero or cannot be bounded, the correct conclusion is threshold dependence, sign indeterminacy, or no quantitative sign claim.

4. Verification and Event Architecture

Verification is layered. A participant can recognize a pointer without reaching it, reach an artifact without matching the intended target, match bytes without interpreting the claim, interpret a claim without agreeing with other participants on the same version, or validate a signature without establishing the truth or authorization of a signed status field.

4.1 Local participant event

$$E_j = R_j \text{ intersect } A_j \text{ intersect } V_j \text{ intersect } I_j \quad (2)$$

Symbol	Meaning	Non-inference
R_j	Participant j recognizes the pointer, identifier, or reference record.	Recognition does not establish availability.
A_j	Participant j can access the required artifact or defensible representation.	Access does not establish identity or historical fidelity.
V_j	Participant j performs the disclosed local verification act.	Local verification need not be target-anchored and does not establish truth.
I_j	Participant j interprets the artifact for the task within the relevant scope.	Interpretation may be self-reported or judgment-dependent.
E_j	All four local conditions hold for participant j.	Local success does not establish cross-participant agreement.

4.2 Cross-participant agreement

V_agree is a separate event: all required participants have verified the same scoped artifact, edition, identifier, digest where applicable, cited claim, and task mapping. Multiple local V_j events can hold while V_agree fails because participants retrieved different editions or mapped the artifact to different claims.

$$\text{Phi_ref} = \text{J_local} \text{ intersect } \text{V_agree} \quad (3)$$

Phi_ref is the truth-anchored target proposition used by the finite-depth epistemic construction.

Phi_ref is fixed before $\text{M_Gamma}^{(d,\tau)}$ is evaluated: J_local and V_agree are defined from the proposition, target, and agreement events rather than from H_d or M_Gamma. Because $\text{H_d}^{\tau}(\text{Phi_ref})$ is truth-anchored, it entails Phi_ref inside the model; the separately named M_Gamma conjunct records the additional task-sufficiency and belief-to-action bridge, not a circular redefinition of the target proposition.

4.3 Coordination outcome partition

$$\text{F_proc} = \text{not } \text{E_1} \text{ union } \dots \text{ union not } \text{E_k} \quad (4)$$

$$\text{D} = (\text{intersection_j } \text{E_j}) \text{ intersect not } \text{V_agree} \quad (5)$$

$$\text{F_mutual} = (\text{intersection_j } \text{E_j}) \text{ intersect } \text{V_agree} \text{ intersect not } \text{M_Gamma}^{(d,\tau)} \quad (6)$$

$$\text{E_coord} = (\text{intersection_j } \text{E_j}) \text{ intersect } \text{V_agree} \text{ intersect } \text{M_Gamma}^{(d,\tau)} \quad (7)$$

$$\text{F_time} = \text{E_coord} \text{ intersect not } \text{T_delta} ; \text{ E_coord,delta} = \text{E_coord} \text{ intersect } \text{T_delta} \quad (8)$$

E_coord is a derived intermediate. At the terminal level, the mutually exclusive partition is {F_proc, D, F_mutual, F_time, E_coord,delta}. The same timing failure may not then be charged again as a second generic coordination failure.

4.4 Artifact and claim support

Object	Question	Typical evidence class
S_artifact(s)	Does source s match the required edition, identifier, digest, and document scope?	Often reproducibly machine-verifiable where targets are encoded.
S_claim(c,s)	Does source s substantively support attributed claim c in context?	Semantic and judgment-dependent.
E_art(s)	Is the intended artifact available, target-matched, and artifact-scope matched?	Composite; machine and dependency checks.
E_support(c,s)	Does the artifact support the claim after scope and semantic review?	Composite; cannot be inferred from byte identity alone.
E_issue	Was the message or status record issued under the assumed key and identity assurances?	Cryptographic and dependency-bound; not truth.

VERIFICATION BOUNDARY Local verification, target-anchored byte verification, cross-participant agreement, artifact matching, claim support, issuance, identity, authorization, truth, relevance, and task value are distinct concepts. A successful check may establish only the field it actually tests.

5. Epistemic Coordination

Common knowledge is a strong benchmark for coordinated action, but practical distributed systems may not attain it. Common p-belief weakens certainty while preserving an infinite doxastic hierarchy. A finite hierarchy can sometimes be task-sufficient, but Rubinstein-type discontinuities show that arbitrarily deep finite belief need not reproduce common-knowledge behavior. The sources therefore support a bounded architecture rather than a universal approximation claim. [R1-R4]

5.1 Cumulative truth-anchored hierarchy

$$H_0^{\tau}(\Phi) = \Phi \quad (9)$$

$$H_{(r+1)}^{\tau}(\Phi) = H_r^{\tau}(\Phi) \text{ intersect } \bigcap_{j=1}^k B_j^{\tau}(H_r^{\tau}(\Phi)) \quad (10)$$

$$C_G^{\tau}(\Phi) = \bigcap_{r=0}^{\infty} H_r^{\tau}(\Phi) \quad (11)$$

$B_j^{\tau}(X)$ is the event that participant j assigns probability at least τ to X . The recursion is cumulative because $H_{(r+1)}$ explicitly retains H_r . C_G^{τ} is a truth-anchored VRP benchmark, not the source-defined doxastic common p-belief object. The notation does not make internal beliefs observable.

5.2 Task-specific finite sufficiency

$$M_{\Gamma}^{(d,\tau)}(\Phi_{\text{ref}}) = H_d^{\tau}(\Phi_{\text{ref}}) \text{ together with disclosed task-sufficiency assumptions } \Gamma \quad (12)$$

The depth d , threshold τ , timing tolerance, action rule, payoff discontinuities, communication reliability, and loss asymmetry are scenario-fixed unless separately calibrated. The paper may conclude that $M_{\Gamma}^{(d,\tau)}$ is sufficient only for the disclosed task model. It may not call the finite condition common p-belief, almost common knowledge, or a general substitute for common knowledge.

5.3 Timing

A group can satisfy the epistemic condition too late. T_{Δ} therefore encodes the task-specific temporal requirement, such as a bounded observation window, deadline, or permitted spread among participant confirmations. Timing may be directly observed even while the underlying belief hierarchy remains latent.

EPISTEMIC NON-CLAIM Public availability, a shared digest, multiple acknowledgments, or repeated citation does not by itself establish higher-order belief, common p-belief, common knowledge, or task-sufficient mutual expectation.

6. Value and Negative Value of Information

Blackwell comparison supplies a narrow gross benchmark: under Bayesian expected-utility optimization, optional use, unchanged actions and objectives, and exclusion of acquisition, processing, commitment, privacy, and third-party costs, a more informative experiment weakly expands the attainable decision set. This does not establish that a public reference is Blackwell-superior, that two real information structures are comparable, or that net or social value is nonnegative. [R5]

6.1 Gross benchmark

The optionality intuition matters. A costless signal can be ignored. A public reference that must be retrieved, interpreted, reviewed, cited, or obeyed cannot be treated as costless merely because the file is publicly reachable. Howard supports decision-specific value and nonadditivity only to the extent verified by the article abstract in the Phase 2 record. [R6]

6.2 Private and social value

$$V_{\text{priv},i} = W_{\text{ref},i} - W_{\text{alt},i} \quad (13)$$

$$V_{\text{soc}} = \sum_i \omega_i (W_{\text{ref},i} - W_{\text{alt},i}) + \Delta W_{\text{external}} \quad (14)$$

Hirshleifer demonstrates that private and social values of information can diverge through timing, redistribution, risk sharing, and dissemination effects. Morris and Shin show that a noisy public signal can operate both as a fundamentals signal and a coordination focal point; under some modeled conditions it can amplify common error or displace private information. The 2006 exchange limits any universal anti-transparency reading. [R7-R10]

6.3 Negative-value mechanisms

Mechanism	Possible accounting location	Required boundary
Process burden	C_proc	Retrieval, verification, interpretation, access, attention, and handling.
Common-mode error	Negative K_coord or a named loss, never both	Specify the shared error and affected population.
Crowding out	K_coord, C_other, or named externality	Identify displaced independent evidence and count once.
Strategic leakage	C_mis or named externality	State what citation reveals about identity, priorities, evidence, or intended action.
Irreversible commitment	C_fail or named loss	Disclose timing and reversibility.
Distributional effects	V_soc weights and externalities	Identify all parties and authorized weights.
Mandatory attention	C_proc and optionality boundary	Do not invoke free-information nonnegativity.

VALUE BOUNDARY Information quantity, uncertainty reduction, citation availability, persistence, prestige, or cryptographic structure is not decision value. Value is relative to actions, consequences, objectives, costs, timing, affected parties, and the alternative branch.

7. Communication and Compression

Shannon supports engineering claims about symbol transmission and source coding while explicitly excluding semantic meaning from the communication problem. Yao supports the task-specific exchanged-information insight at the level verified by the 1979 abstract. Slepian-Wolf and index-coding results show that side information and shared broadcast can reduce rates under precise distribution, decoder, sender-knowledge, and fidelity assumptions. None establishes that a short citation is understood, correct, useful, or cheap to integrate. [R11-R15]

7.1 Shared and marginal transmission

$$C_{\text{tx},b,\Gamma}(J) = C_{\text{tx},b,\Gamma}^{\text{shared}}(J) + \sum_{j \in J} C_{\text{tx},b,j,\Gamma}^{\text{marg}} \quad (15)$$

The sender may pay one shared broadcast cost while recipients face heterogeneous marginal formatting, authentication, or privacy costs. Shared transmission does not imply shared side information or shared interpretation.

7.2 Reference branch transmission and fallback

$$E_{\Gamma}[C_{\text{tx},\text{ref}}(J)] = C_{\text{ptr},\text{shared},\Gamma}(J) + \sum_j [C_{\text{ptr},j,\text{marg},\Gamma} + p_{\text{fail},j,\Gamma} C_{\text{fallback},\text{tx},j,\Gamma}] \quad (16)$$

$$K_{\text{comp},\Gamma}(J) = E_{\Gamma}[C_{\text{tx},\text{alt}}(J)] - E_{\Gamma}[C_{\text{tx},\text{ref}}(J)] \quad (17)$$

K_{comp} is signed. It is positive when expected sender/channel transmission is lower in the reference branch, negative when reference transmission and fallback exceed the alternative, and zero when they are equal. Fallback belongs inside the reference branch before K_{comp} is computed. Failure followed by the ordinary alternative is not a second compression loss.

7.3 Optional special-case identity

$$K_{comp, \Gamma(J)} = \sum_j p_{use,j} G_{comp,j} + \sum_j p_{fail,j} [C_{tx,alt,j} - C_{fallback,tx,j}] - C_{ptr,shared} - \sum_j C_{ptr,j,marg} \quad (18)$$

Legacy Equation 8a. It is subordinate to the general both-branch Equation 17 in this paper.

Equation 18 may be used only when all four validity conditions hold: $G_{comp,j}$ equals the per-recipient alternative transmission cost; $p_{use,j} + p_{fail,j} = 1$ is an exhaustive binary partition; alternative transmission is recipient-additive with no shared alternative component; and the successful reference path transmits no payload beyond the displayed pointer terms. Otherwise the general both-branch definition in Equation 17 controls.

7.4 Recipient processing

$$C_{recv,j} = C_{resolve,j} + C_{ret,j} + C_{dec,j} \quad (19)$$

$$C_{proc,j} = C_{access,j} + C_{recv,j} + C_{ver,j} + C_{int,j} \quad (20)$$

Economic access, receiving, retrieval, decoding, verification, interpretation, and handling remain outside K_{comp} . C_{ptr} is a sender/channel transmission input, not a terminal cost address. C_{delay} is a routing-only diagnostic label: each concrete delay is assigned once to K_{comp} , C_{proc} , C_{fail} , or a named externality according to its effect, and is never charged separately after routing.

7.5 Fixed-size pointer boundary

$$ell_{ptr}(N; \lambda, \text{scheme}) = O_{(\lambda, \text{scheme})}(1) \text{ relative to payload length } N; \quad C_{ref, \Gamma(N,J)} \text{ need not be } O(1) \quad (21)$$

NO HASH-TO-COMPRESSSION SHORTCUT A digest may be compact relative to payload length under a fixed scheme and security parameter. It does not convey the object, prove possession, preserve context, establish semantic scope, or eliminate retrieval, verification, interpretation, and fallback.

8. Provenance and Audit Channels

Provenance and audit legibility are separate signed channels. Provenance concerns whether a reference reduces disputes about origin, edition, version, or artifact identity. Audit legibility concerns whether an authorized reviewer can reconstruct and classify the decision process more accurately. Neither channel proves the substantive truth or authority of the cited claim.

8.1 Provenance channel

$$E_{prov} = R \text{ intersect } A \text{ intersect } V_{target} \quad (22)$$

$$K_{prov} = P(E_{prov}) \Delta p_{disp|E_{prov}} L_{prov} \quad (23)$$

A positive provenance contribution requires an identified dispute class, a target-anchored verification condition, a disclosed alternative dispute probability, and a loss magnitude. If a hash target is untrusted, the bytes are missing,

the wrong edition is retrieved, or the claim falls outside the artifact scope, provenance value may be zero or negative after costs.

8.2 Audit channel

$$E_{\text{aud}} = R_{\text{aud}} \text{ intersect } A_{\text{aud}} \text{ intersect } V_{\text{aud}} \text{ intersect } I_{\text{aud}} \quad (24)$$

$$K_{\text{audit}} = P(E_{\text{aud}}) \Delta p_{\text{misclass}} | E_{\text{aud}} L_m \quad (25)$$

The authorized auditor set, review criteria, evidence access, classification target, and loss of misclassification must be disclosed. Public readability is not identical to auditability, and auditability is not identical to authority. Corrective audit labor is a process cost even when the audit channel has positive expected value.

8.3 Signed-channel convention

$$\Delta p_x = p_{x,\text{alt}} - p_{x,\text{ref}} \text{ in } [-1,1] \quad (26)$$

A positive Δp_x means the reference branch reduces the specified undesirable outcome relative to the alternative.

FOUR-CHANNEL LIMIT The intrinsic signed channels are exactly K_{coord} , K_{comp} , K_{prov} , and K_{audit} . Durability and strategic behavior condition these channels or enter costs, losses, and externalities; they do not create a fifth positive channel.

9. Persistence and Durability

A persistent name is not a permanently available object. DOI infrastructure separates identifier names from maintained records and resolving locations. Hash-derived names bind retrieved bytes to a trusted digest but do not locate missing content. Memento supports temporal discovery but cannot guarantee a representative capture. Historical studies show link rot and content drift in specified corpora, not universal current decay constants. Robust Links preserve original URI, linking time, and snapshot pathways as an experimental mitigation pattern, not a permanence guarantee. [R16-R22]

9.1 Usable path event

$$E_{m,t} = A_{m,t} \text{ intersect } V_{m,t}^{\text{target}} \text{ intersect } S_{m,t} \quad (27)$$

$A_{m,t}$ is practical accessibility; $V_{m,t}^{\text{target}}$ is target-anchored byte verification where an exact target exists; $S_{m,t}$ is edition, identifier, scope, and metadata agreement. When no exact target digest exists, timestamp, provenance, or similarity evidence must be labeled weaker than byte verification.

9.2 Correlation-aware redundancy

$$p_{\text{usable},t} = P_{\text{Gamma}}(\text{union}_m E_{m,t}) \quad (28)$$

The product formula $1 - \text{product}_m(1 - p_{m,t})$ is permitted only under a defended independence assumption. Common gateways, resolvers, providers, credentials, operators, metadata services, legal constraints, and software stacks create dependence domains. Nominal copy count is not independent survival probability.

9.3 Recurring preservation vector

$$C_{\text{pres},b,t} = C_{\text{store},b,t} + C_{\text{meta},b,t} + C_{\text{monitor},b,t} + C_{\text{migrate},b,t} + C_{\text{pres,verify},b,t} + C_{\text{repair},b,t} \quad (29)$$

Per-use verification $C_{\text{ver,event}}$ remains separate from preservation-level re-verification $C_{\text{pres,verify}}$. Shared archive costs and item-level marginal costs must be disclosed. Detected drift routed to repair cost may not also be charged as realized drift harm.

9.4 Decay losses

$$L_{\text{decay},t}^{\text{exp}} = P_{\text{Gamma}}(\text{not usable at } t) L_{\text{unavailable},t} + P_{\text{Gamma}}(\text{drift at } t \text{ and used}) L_{\text{drift},t} \quad (30)$$

$L_{\text{unavailable},t}$ routes to C_{fail} by default; $L_{\text{drift},t}$ routes to C_{div} by default. An alternative route requires scenario-specific justification and one-charge reconciliation. Accessibility and content fidelity are separate observations: a live response can still deliver materially changed content.

DURABILITY BOUNDARY Durability is a time-indexed eligibility, failure-risk, and recurring-cost layer. Intended permanence, a DOI, a hash, a snapshot, an Arweave transaction, or multiple mirrors does not guarantee future retrieval, historical fidelity, interpretation, or authority.

10. Strategic Reference Behavior

A sender may cite to inform, coordinate, persuade, signal diligence, transfer prestige, conceal missing evidence, inflate verification status, or reveal operational information. Strategic communication can remain partially informative under preference conflict; costly signals can separate types only under incentive-compatibility conditions; lying costs need not eliminate inflated language; and verifiable disclosure results depend on detectability, completeness, receiver inference, and cost. Digital signatures establish integrity and signatory-assurance functions under their assumptions, not truth, relevance, completeness, or authorization. [R23-R30]

10.1 Layered message

$$m = (r, c, z, \text{sigma}, D) \quad (31)$$

Field	Meaning	Boundary
r	Pointer or artifact identifier.	A working pointer does not establish claim support.
c	Claim attributed to the source.	Must be tested through $S_{\text{claim}}(c,s)$.
z	Status vector, such as verified, reviewed, authorized, or final.	Each status field needs its own evidence.
sigma	Signature or issuance evidence.	Valid issuance is not truth or authorization.
D	Disclosed evidence set.	Completeness and material omissions may be judgment-dependent.

10.2 Strategic events

Event	Definition or test	Accounting effect
F_status	A material verification or authorization status is false, inflated, or unsupported.	C_{mis} , C_{fail} , or named externality.
T_cit	Citation form or prestige is used as theater without task support.	Process burden and misrepresentation loss.
S_select	Material evidence is withheld and full evidence would alter receiver action.	C_{mis} , C_{fail} , or externality, once.
U_authority	The receiver is invited to infer unsupported authority from the cited artifact.	Judgment-dependent condition for laundering.
A_launder	Authentic artifact, no claim support, and unsupported authority inference.	Strategic distortion, not provenance value.

$$A_launder(c,s) = E_art(s) \text{ intersect not } S_claim(c,s) \text{ intersect } U_authority(c,s) \quad (32)$$

$$C_mis, \Gamma = \sum_h P_Gamma(F_h) L_h \quad (33)$$

The indexed harms F_h must be materially distinct or reconciled before aggregation.

10.3 Costly-signal boundary

A costly archive deposit, signature ceremony, long bibliography, blockchain transaction, or preservation expense can affect beliefs only under disclosed type-dependent costs and benefits. Expenditure alone is not truth. If low-quality or deceptive types can imitate the signal profitably, separation collapses.

10.4 Strategic leakage

The act of citing may reveal identity, priorities, evidence state, operational scope, intended action, or confidence. These effects can be positive, negative, or strategically exploited. They belong in the relevant signed channel, C_mis , C_fail , or a named externality under the one-charge rule.

11. Integrated Static and Dynamic Premium

11.1 Absolute branch primitives

$$W_{b,\Gamma} = E_{\Gamma}[U_{task,b} - \Lambda_{coord,b,\Gamma} - \Lambda_{prov,b,\Gamma} - \Lambda_{audit,b,\Gamma} - C_{tx,b,\Gamma} - \sum_j C_{proc,b,j,\Gamma} - C_{fail,b,\Gamma} - C_{div,b,\Gamma} - C_{mis,b,\Gamma} - C_{other,b,\Gamma}] \quad (34)$$

$\Lambda_{coord,b}$, $\Lambda_{prov,b}$, and $\Lambda_{audit,b}$ are branch-level expected loss primitives. $C_{tx,b}$ is absolute sender/channel transmission cost. The four K channels are cross-branch differences, not branch-indexed addends: for example, $K_{coord} = E[\Lambda_{coord,alt} - \Lambda_{coord,ref}]$ and $K_{comp} = E[C_{tx,alt} - C_{tx,ref}]$.

$$P_{i,ref,\Gamma} = W_{ref,\Gamma} - W_{alt,\Gamma} \quad (35)$$

11.2 Decomposed reconciliation

$$P_{i,ref,\Gamma} = \Delta U_{task,\Gamma} + K_{coord,\Gamma} + K_{comp,\Gamma} + K_{prov,\Gamma} + K_{audit,\Gamma} - \Delta C_{tot,\Gamma} \quad (36)$$

ΔU_{task} is the expected task-utility difference between branches. ΔC_{tot} is the net difference across non-transmission terminal costs and named externalities. Equation 36 must reconcile exactly with the two absolute values in Equation 34. A one-sided shortcut is permitted only when the omitted alternative-side quantity is explicitly zero under Γ .

11.3 Dynamic premium

$$K_{ref,t} = K_{coord,t} + K_{comp,t} + K_{prov,t} + K_{audit,t} \quad (37)$$

$$P_{i,ref,\Gamma}^{dyn}(T) = -\Delta C_{setup} + \sum_{(t=0)^T} \beta^t E_{\Gamma}[\Delta U_{task,t} + K_{ref,t} - \Delta C_{tot,t} - \Delta C_{pres,t}] \quad (38)$$

$K_{ref,t}$ is shorthand for the four-channel sum, not a fifth channel. $\Delta C_{tot,t}$ expressly excludes preservation components charged through $\Delta C_{pres,t}$. Setup and preservation terms are both-sided differentials. A one-sided form is valid only when the corresponding alternative-side setup or preservation cost is explicitly zero. For a finite horizon, $0 \leq \beta \leq 1$. For an infinite horizon, $\beta < 1$ is necessary but not sufficient; the discounted expected sequence must converge.

11.4 One-charge rule

ONE-CHARGE RULE A concrete unit of traffic, labor, delay, ambiguity, crowd-out, disclosure leakage, fallback, divergence, misrepresentation, or external harm receives exactly one accounting address in a given application. Events may overlap. Concrete harms may not be duplicated.

Terminal address	Contents	Exclusion
U_task	Direct authorized task utility.	Do not hide channel value or externalities here.
K_coord	Cross-branch coordination-loss difference.	No duplicate common-mode or timing harm.
K_comp	Cross-branch sender/channel transmission difference.	No recipient retrieval, verification, or interpretation.
K_prov	Cross-branch origin/version dispute-loss difference.	No truth or authority inference.
K_audit	Cross-branch authorized audit-misclassification difference.	Corrective labor remains C_proc.
C_proc	Access, receiving, verification, interpretation, and handling.	No sender transmission already in K_comp.
C_fail	Incremental task or decision failure consequence.	Do not repeat process effort or transmission.
C_div	Version, edition, temporal, or scope divergence loss.	Do not repeat drift repair cost.
C_mis	Misrepresentation, false status, unsupported reliance, or strategic distortion.	Harms must be disjointly defined or reconciled.
C_pres	Storage, metadata, monitoring, migration, preservation verification, and repair.	Separate from per-use verification.
C_other	Residual named cost not already represented.	May not absorb a specified term.

12. Identification, Calibration, and Uncertainty

MANDATORY TECHNICAL SPINE

Formal notation does not establish observability, empirical identification, calibration, or normative authority. Before the first numerical premium, every input must be classified across evidence status, quantification form, calibration basis, dependencies, accounting address, and the mandatory reporting rule $R(x)$, including unresolved status and reason where applicable.

$$L(x) = [I(x), Q(x), C(x), D(x), A(x), R(x)] \quad (39)$$

Identification status, quantification form, calibration basis, dependencies, accounting/authority record, and mandatory reporting rule for input x.

$R(x)$ states the disclosure required whenever x is used, including its evidence boundary, unresolved status, prohibited inference, and any scenario or special-case condition. In the companion schema, disclosure_note and related boundary fields implement the machine-readable portion of $R(x)$; they do not remove the formal reporting field from the ledger.

12.1 Identification axis

Code	Class	Publication treatment
I1	Directly observed	State instrument or system, unit, time, scope, missingness, and measurement limits.
I2	Reproducibly machine-verifiable	State target, algorithm, key material,

Code	Class	Publication treatment
		environment, and what remains unproven.
I3	Externally verifiable with dependencies	Name registry, authority, identity chain, policy, and failure dependencies.
I4	Historically estimated	Report corpus, period, estimator, interval, transfer limits, and drift.
I5	Model-inferred or model-estimated	State model, priors, dependence, alternatives, fit limits, and sensitivity.
I6	Self-reported or asserted	Preserve speaker, wording, time, scope, and corroboration status.
I7	Judgment-dependent	State reviewer, criteria, evidence set, disagreement procedure, and contestability.
I8	Normative or authority-fixed	Identify authorized objective, weights, loss severity, and participant scope.
I9	Scenario-fixed assumption	Label as a modeling choice and vary adequacy in sensitivity analysis.
I10	Partially identified or bounded	Report admissible set and assumptions supporting the bounds.
I11	Unidentified	Do not assign false precision; use scenarios, thresholds, or qualitative conclusions.

12.2 Quantification and calibration

Axis	Codes
Q - quantification	Q1 point; Q2 interval; Q3 distributional; Q4 set-valued; Q5 scenario-valued; Q6 qualitative ordered; Q7 unquantified.
C - calibration	C1 direct measurement; C2 controlled test; C3 historical estimate; C4 model estimate; C5 expert or reviewer judgment; C6 authorized normative choice; C7 scenario assumption; C8 illustrative unit; C9 uncalibrated.

Identification and calibration are separate. A model may assign an illustrative value to an unidentified parameter for sensitivity analysis without converting it into an empirical estimate. An observed value may still be noisy or interval-valued. A judgment-dependent event may be represented by a scenario set without becoming machine-verifiable.

12.3 Robust sign

Θ_{Γ} = disclosed admissible set of parameter values and dependence structures (40)

$$P_{\text{lower}} = \inf_{(\theta \in \Theta_{\Gamma})} P_{\text{ref}, \Gamma}(\theta); \quad P_{\text{upper}} = \sup_{(\theta \in \Theta_{\Gamma})} P_{\text{ref}, \Gamma}(\theta) \quad (41)$$

Condition	Permitted conclusion
$P_{\text{lower}} > 0$	Robustly positive over the disclosed admissible set.
$P_{\text{upper}} < 0$	Robustly negative over the disclosed admissible set.
$P_{\text{lower}} \leq 0 \leq P_{\text{upper}}$	Sign-indeterminate; report the reversal drivers or threshold.
No defensible finite bounds	No quantitative sign claim; retain qualitative or threshold conclusions.

12.4 No zero by omission

An omitted externality, unknown belief state, hidden evidence set, unmeasured maintenance burden, or unresolved dependence is not zero. It must be named, bounded, varied, or left explicitly unquantified. Every worked-case number below is illustrative-only unless otherwise stated; no unit is a dollar, probability estimate, prevalence rate, or empirical calibration.

13. Worked Cases

The cases locate mechanisms and test accounting. They do not estimate population prevalence, typical archive value, or future artificial-system behavior. Each case fixes Gamma and Omega. When fixed point-valued branch totals are asserted, both absolute branches are displayed before the premium. For robust-sign analysis in which the branches vary over an admissible uncertainty set, the branch functions and resulting premium bounds may be displayed instead, provided the exception and baseline are explicit. All values are illustrative-only, and every case applies the one-charge rule.

CASE A: Finite-depth coordination can be positive under a bounded task

Shows that a task-specific finite condition may support a positive premium without being called common knowledge.

Gamma_A: three participants must select the same reversible configuration within ten minutes; $d = 2$, $\tau = 0.80$, and the loss from mismatch is moderate. Omega_A contains the same three participants and evidence state before branch assignment. Values are illustrative-only (I9/Q5/C8).

Row	Reference	Alternative	Address
Task utility	120	120	U_task
Expected coordination loss	-6	-26	Lambda_coord
Transmission	-8	-18	C_tx
Recipient process	-12	-7	C_proc
Absolute value	94	69	W_b

$$Pi_{ref,A} = 94 - 69 = +25 \text{ illustrative units} \quad (42)$$

The implied channel differences are $K_{coord} = 20$ and $K_{comp} = 10$, offset by five additional process units. The sign depends on the assumed task sufficiency of $d = 2$ and the timing window; neither internal belief nor the behavioral bridge is directly observed.

CASE B: Large finite depth does not guarantee common-knowledge behavior

Rubinstein-style boundary: many finite confirmations may fail to change equilibrium behavior in a discontinuous task.

Gamma_B: a high-loss irreversible action is optimal only under common knowledge of a state. The reference branch creates a long but finite confirmation chain; the alternative branch creates none. The task model retains the safe action in both branches because any finite chain leaves a strategically decisive residual doubt. [R4]

Row	Reference	Alternative
Observable confirmations	Large finite number	None
Action under disclosed equilibrium model	Safe action	Safe action
Absolute expected value	40	40
Hypothetical common-knowledge benchmark	90 (not a branch)	90 (not a branch)

$$Pi_{ref,B} = 40 - 40 = 0 \text{ illustrative units} \quad (43)$$

The case blocks a universal positive K_{coord} inference from finite depth. The common-knowledge benchmark is shown only to explain the discontinuity; it is not silently substituted for the alternative branch.

CASE C: Gross information gain can coexist with negative net value

Separates the ideal optional-information benchmark from complete branch accounting and social effects.

Gamma_C: a noisy public anchor improves gross fundamental matching but creates processing burden, common-mode error, and crowding out of independent signals. Omega_C is the same affected group in both branches. Values are illustrative-only.

Row	Reference	Alternative	Address
Gross task value	130	100	U_task
Process cost	-12	-4	C_proc
Common-mode error	-26	0	Lambda_coord,ref; $K_{\text{coord}} = -26$
Crowd-out externality	-10	0	Named externality
Absolute value	82	96	W_b

$$Pi_{\text{ref},C} = 82 - 96 = -14 \text{ illustrative units} \quad (44)$$

The reference has a 30-unit gross advantage but a 14-unit negative net premium. The example is an existence construction, not evidence that public information is generally harmful.

CASE D: Shared transmission can create a positive multi-recipient premium

Shows genuine broadcast savings while retaining recipient-specific process costs.

Gamma_D: five recipients require the same task-relevant evidence. The sender knows the recipients and can use one shared pointer plus one marginal unit per recipient. The alternative sends separate 24-unit task packets. Recipient-side work remains heterogeneous and nonzero.

Row	Reference	Alternative	Address
Task utility	220	220	U_task
Shared transmission	-8	0	C_tx shared
Marginal transmission	-5	-120	C_tx marginal
Recipient process	-60	-25	C_proc
Absolute value	147	75	W_b

$$K_{\text{comp},D} = 120 - 13 = +107 ; \quad Pi_{\text{ref},D} = 147 - 75 = +72 \text{ illustrative units} \quad (45)$$

Transmission savings exceed the additional 35 units of recipient processing. The case does not infer that the pointer is understood, that side information is current, or that every recipient can retrieve the artifact.

CASE E: A valid hash name with no retrievable copy can be negative

Shows pointer and fallback overshoot without treating byte identity as availability.

Gamma_E: the digest correctly names target bytes, but no usable copy is retrievable before the decision window. Fallback eventually sends the same full content used by the alternative. Values reproduce the confirmed durability-stream illustration and remain illustrative-only.

Row	Reference	Alternative	Address
Task utility after fallback	60	60	U_task
Pointer transmission	-1	0	C_tx
Failed resolve/retrieval effort	-6	0	C_proc

Row	Reference	Alternative	Address
Fallback full transmission	-15	-15	C_tx
Missed-window consequence	-10	0	C_fail
Absolute value	28	45	W_b

$$P_{i_ref,E} = 28 - 45 = -17 \text{ illustrative units} \quad (46)$$

Fallback equals the alternative and therefore cancels in the difference. The negative premium is exactly the pointer, failed effort, and missed-window consequence. The zero alternative-side failed-retrieval effort is scenario-assumed because the alternative begins with direct full transmission and performs no pointer resolution; it is not a general property of alternative branches. No separate generic “reference failure” charge is added.

CASE F: Target-anchored citation creates a positive strategic-reference premium

Carry-forward of the confirmed v0.5.2 case: positive value comes from actual support and auditability, not prestige.

Primitive restatement note: to conform to Equation 34, the confirmed +8 provenance and +6 audit channel contributions are represented as avoided alternative-branch loss primitives. The confirmed absolute totals and +19 premium are unchanged.

Item	Reference	Alternative	Address
Common task utility baseline	114	114	U_task
Expected provenance-dispute loss	0	-8	Lambda_prov
Expected audit-misclassification loss	0	-6	Lambda_audit
Process and verification	-5	-7	C_proc
Residual misclassification	-2	0	C_mis
Unresolved dispute	0	-5	C_fail
Absolute value	107	88	W_b

$$K_{prov,F} = 8 ; K_{audit,F} = 6 ; P_{i_ref,F} = 107 - 88 = +19 \text{ illustrative units} \quad (47)$$

The reference branch is positive because the retrieved artifact is target-matched, actually supports the attributed claim, and reduces expected provenance dispute and authorized audit misclassification after processing and residual error are counted.

CASE G: Citation theater can be negative despite transmission saving

Demonstrates that positive K_{comp} and total reference value are independent quantities.

Item	Reference	Alternative	Address
Task utility	100	100	U_task
Sender/channel transmission	-4	-6	Determines $K_{comp} = +2$
Recipient processing and review	-6	-6	C_proc
Unsupported-reliance loss	-18	0	C_mis
Absolute value	72	88	W_b

$$K_{comp,G} = 6 - 4 = +2 ; P_{i_ref,G} = 72 - 88 = -16 \text{ illustrative units} \quad (48)$$

The pointer works and saves two transmission units, but S_claim is false and the receiver relies on an unsupported proposition. C_ptr is not a terminal address; transmission determines K_comp while recipient labor remains C_proc .

CASE H: A valid signature can accompany a false status field

Separates issuance integrity from truth of the signed verification claim.

Item	Reference	Alternative	Address
Task utility before error	80	80	U_task
Signed-record generation and review	-6	-10	C_proc
Action loss from false verified status	-24	0	C_mis
Absolute value	50	70	W_b

$$Pi_ref,H = 50 - 70 = -20 \text{ illustrative units } (49)$$

Signature validation establishes issuance under the assumed key assurances. It does not make the signed status field true. The 24-unit action loss is charged once to C_mis .

CASE I: Selective disclosure and authority laundering

Shows overlapping events with distinct harms and one-charge reconciliation.

Primitive restatement note: the confirmed +4 provenance contribution is represented as four units of avoided alternative-branch provenance-dispute loss. The confirmed absolute totals and -17 premium are unchanged.

Item	Reference	Alternative	Address
Common task utility baseline	104	104	U_task
Expected provenance-dispute loss	0	-4	Λ_prov
Initial processing burden	-5	-12	C_proc
Broader decision distortion	-22	0	C_mis
Corrective audit labor	-6	0	C_proc
Absolute value	71	88	W_b

$$K_prov,I = 4 ; Pi_ref,I = 71 - 88 = -17 \text{ illustrative units } (50)$$

Step	Controlled reconciliation
1. Root artifact	$E_art(s)$ holds: the source is accessible, target-verified, and artifact-scope matched.
2. Selective disclosure	S_select holds because D omits material evidence and full evidence would change the action.
3. Authority laundering	$A_launder$ holds because $S_claim(c,s)$ is false while $U_authority$ transfers source prestige to the broader claim.
4. Genuine narrow value	The four-unit provenance improvement is represented once as avoided alternative-branch Λ_prov loss, yielding $K_prov = +4$.
5. One-charge routing	-22 goes once to C_mis ; -5 and -6 are distinct process labor in C_proc . No harm is repeated.

CASE J: Costly signaling separates only above a type-dependent threshold

Shows why expense is not automatic credibility and how separation collapses when imitation becomes affordable.

Assume equal acceptance benefit $\Delta U_H = \Delta U_L = 8$, high-type signal cost $c_H(s) = 3$, and low-type cost $c_L(s) = 12$.

Type	Signal payoff	Result
H	$8 - 3 = +5$	Signaling is privately worthwhile.
L	$8 - 12 = -4$	Imitation is not worthwhile; separation is possible.
L after cost drop	$8 - 5 = +3$	Both types signal; separation collapses.

This is an incentive-compatibility illustration, not evidence that expensive citation infrastructure is credible in practice. Signal cost, acceptance benefit, type distribution, imitation technology, and independent productivity remain unidentified unless measured.

CASE K: Dynamic durability premium changes sign under recurring-cost sensitivity

Shows that persistence value is conditional on maintenance effectiveness and cost.

Gamma_K: five periods, $\beta = 0.90$, setup differential 12, and a base per-period reference advantage of 5 after process and preservation costs. The dynamic $\Delta C_{tot,t}$ used here excludes preservation costs, which are represented only through $\Delta C_{pres,t}$.

$$\Pi_{ref,K}^{dyn} = -12 + 5(1 + 0.9 + 0.81 + 0.729 + 0.6561) = +8.4755 \quad (51)$$

If recurring preservation and repair costs rise by three units per period, the net per-period advantage falls to two:

$$\Pi_{ref,K}^{dyn} = -12 + 2(1 + 0.9 + 0.81 + 0.729 + 0.6561) = -3.8098 \quad (52)$$

The same reference design changes sign. Intended permanence is not a permanent positive property; it is a dynamic branch result under stated costs, effectiveness, horizon, and discounting.

CASE L: One root failure can create multiple distinct harms without double counting

Demonstrates event overlap versus duplicated harm.

Gamma_L: reference retrieval fails, forcing fallback and causing a missed deadline. The retrieval failure is a root event, not itself a terminal cost. Its distinct consequences are routed once.

Item	Reference	Alternative	Address
Task utility	100	100	U_task
Pointer transmission	-2	0	C_tx
Fallback transmission	-12	-12	C_tx
Failed retrieval labor	-4	-2	C_proc
Missed-window consequence	-20	0	C_fail
Absolute value	62	86	W_b

$$\Pi_{ref,L} = 62 - 86 = -24 \text{ illustrative units} \quad (53)$$

Fallback transmission cancels. The incremental harms are the pointer (2), extra retrieval labor (2), and missed-window consequence (20). No generic failure cost is added on top. Event overlap is permitted; duplicated concrete harm is not.

CASE M: A positive baseline can remain sign-unidentified

Applies the robust-sign rule when unobserved strategic and external effects cross zero.

Gamma_M yields a baseline premium of +12 from observed and scenario-fixed components. Hidden evidence completeness, authority inference, correlated failure, and external effects jointly contribute an admissible interval from -21 to +19.

$$Pi_{ref,M} \text{ in } 12 + [-21, +19] = [-9, +31] \quad (54)$$

Robust-sign branch-display qualification. Case M evaluates uncertainty-indexed branch functions $W_{ref,M}(\theta)$ and $W_{alt,M}(\theta)$, with $Pi_{ref,M}(\theta) = W_{ref,M}(\theta) - W_{alt,M}(\theta)$ for θ in Θ_M . At the stated baseline parameter vector θ_0 , the premium is +12. Across the admissible set, the premium lies in $[-9, +31]$. Because the underlying branches are uncertainty-indexed rather than fixed scalar totals, this case reports the branch functions and premium bounds directly and does not assert unique absolute scalar values for $W_{ref,M}$ and $W_{alt,M}$.

Because the interval crosses zero, the paper may not report a positive premium. The correct conclusion is sign indeterminacy. The next analytical question is which parameters or thresholds drive the reversal, not which favorable point estimate should be selected.

14. Sensitivity and Robustness

A reference-premium estimate is incomplete without joint sensitivity. Varying one parameter at a time can conceal correlated failure, strategic interaction, selection, and threshold effects. The controlling registry below preserves the exact S1-S10 meanings from the Integration Package v0.1.1; the labels are therefore stable across the prose, appendices, schema records, and worked examples.

Registry family	Axes to vary
S1 Baseline and selection	Omega, alternative branch, optional versus mandatory use, affected-party scope, selection into visible or successful cases.
S2 Epistemic sufficiency	Depth d, thresholds tau or tau_j, timing window, message failure, payoff discontinuity, unilateral-reliance loss, and belief-behavior mapping.
S3 Information and welfare	Public-anchor accuracy, private-signal quality and diversity, crowd-out, coordination motive, common-mode loss, welfare weights, and reversibility.
S4 Communication topology	Recipient count, broadcast, multicast, point-to-point delivery, pointer and metadata size, payload scope, side information, caches, format support, and fallback.
S5 Process and failure burden	Resolution, retrieval, decoding, verification, interpretation, audit capacity, delay, correction, and process cost from near zero to prohibitive. Repair or correction is controlled here when the same burden also appears in S6.
S6 Durability and dependence	Horizon, beta, demand frequency, link, content, and resolver drift, archive sparsity, mirror correlation, access restrictions, migration, and repair. A shared repair burden is varied once and cross-referenced to S5.
S7 Strategic behavior	Preference alignment, lying and signal-cost differentials, omission detection, disclosure cost, source flooding, consensus simulation, and receiver skepticism.
S8 Identity and authority dependencies	Key compromise, delegated signing, identity binding, timestamp authority, authorization disputes, and repository or operator identity.
S9 Leakage and adversarial adaptation	Pointer, metadata, recipient list, access pattern, evidence-state disclosure, adaptive misuse, and privacy or security externalities.
S10 Calibration and model uncertainty	Historical transfer, model form, priors, dependence, reviewer disagreement, normative authority, interval width, and unidentified parameters.

The tables below are illustrative stress tests, not empirical calibrations. They are included to deliver the sensitivity outputs required by the controlling integration package. Each stressed quantity remains scenario-fixed unless separately measured, and every added loss is assigned once.

14.1 One-way sign thresholds

Scenario and varied quantity	Premium expression	Zero or robustness threshold	Baseline conclusion
Case A: additional reference-side process differential q_A	$Pi_A = 30 - q_A$	$q_A = 30$	$q_A = 5$ gives +25; above 30 the sign is negative.
Case C: gross task advantage g_C	$Pi_C = g_C - 44$	$g_C = 44$	$g_C = 30$ gives -14; gross value must exceed 44 to offset the disclosed burdens.
Case D: additional recipient-processing differential q_D	$Pi_D = 107 - q_D$	$q_D = 107$	$q_D = 35$ gives +72; process burden above 107 reverses the sign.
Case F: residual misclassification loss m_F	$Pi_F = 21 - m_F$	$m_F = 21$	$m_F = 2$ gives +19; residual loss at or above 21 removes the positive premium.
Case K: net recurring advantage a_K	$Pi_K = -12 + 4.0951 a_K$	a_K approximately 2.9303	$a_K = 5$ gives +8.4755; $a_K = 2$ gives -3.8098.
Case M: baseline b_M with hidden interval [-21,+19]	Pi_M in $[b_M - 21, b_M + 19]$	Robust positive only if $b_M > 21$; robust negative only if $b_M < -19$	$b_M = 12$ yields [-9,+31], so the sign is unidentified.

These thresholds are properties of the disclosed numerical constructions, not universal constants. A different Omega, alternative branch, objective, or dependence model changes them.

14.2 Per-family stress-test matrix

Family	Illustrative joint stress	Result	Interpretation
S1	Include 20 inaccessible participants omitted by a visible-user-only sample.	Selected sample +10; full Omega -4.	Selection into successful cases can reverse the sign; see Section 14.4.
S2	Weaken the task-sufficiency bridge and tighten timing so Case A K_{coord} falls from +20 to -6.	$Pi_A = -6 + 10 - 5 = -1$.	Finite depth is valuable only through a disclosed task and timing bridge.
S3	Raise Case C gross advantage from 30 to 45 while reducing crowd-out from 10 to 5.	$Pi_C = 45 - 26 - 8 - 5 = +6$.	Information accuracy and welfare externalities jointly determine net sign.
S4	Increase Case D reference transmission from 13 to 85 through larger metadata and weaker side information, with alternative traffic fixed at 120.	$K_{comp} = 35$; with $q_D = 35$, $Pi_D = 0$.	Broadcast advantage disappears at the topology threshold.
S5	Raise Case D additional recipient-processing and failure burden from 35 to 120.	$Pi_D = 107 - 120 = -13$.	Sender savings do not cap total recipient-side burden.
S6	Raise recurring preservation and repair burden so Case K net recurring advantage falls from 5 to 2 at $\beta = 0.90$.	Dynamic premium changes from +8.4755 to -3.8098.	Durability value is conditional on recurring effectiveness, cost, horizon, and dependence.
S7	Use the authentic but non-supporting citation in Case G with receiver reliance and an 18-unit distortion loss.	$K_{comp} = +2$ but $Pi_G = -16$.	Strategic misuse can dominate a real transmission saving.
S8	Add one 20-unit identity or authorization dependency loss to otherwise positive Case F.	$Pi_F = +19 - 20 = -1$.	Issuance and identity dependencies can reverse a provenance advantage; the loss is charged once.
S9	Add a 24-unit privacy or adaptive-misuse externality to Case F because the pointer and access pattern reveal the evidence state.	$Pi_F = +19 - 24 = -5$.	Reference visibility can create adversarial or privacy costs outside the four positive channels.
S10	Widen jointly uncertain strategic, dependence, and model-form terms as in Case M.	Pi_M in [-9,+31].	Reviewer disagreement or model uncertainty that crosses zero requires sign indeterminacy.

The S5 and S6 rows cross-reference one repair mechanism rather than charging or varying it twice. S8 and S9 use named additional losses only for the stressed scenario; they do not create new positive channels.

14.3 Dependence stress test

Suppose three nominal mirrors each have a disclosed 0.10 failure probability, a complete retrieval failure produces a 20-unit missed-window loss, and the reference branch has a +1 premium before availability loss. The numbers are illustrative-only and compare dependence assumptions, not actual repository reliability.

Dependence model	Probability all retrieval paths fail	Expected missed-window loss	Resulting premium
Independent failures	$0.10^3 = 0.001$	$20 \times 0.001 = 0.02$	$+1 - 0.02 = +0.98$
Perfect common-mode failure	0.10	$20 \times 0.10 = 2.00$	$+1 - 2.00 = -1.00$

Counting three URLs without testing provider, gateway, credential, resolver, metadata, operator, or legal dependence can therefore produce a false robust-positive conclusion. The model must vary plausible joint failure structure rather than factor probabilities by convenience.

14.4 Selection-bias check

A visible-user-only analysis conditions on reference success after the branch split. The following construction restores the common pre-branch Omega of 100 participants. Eighty can use the reference and have $W_{ref} = 110$ versus $W_{alt} = 100$. Twenty encounter an access barrier and have $W_{ref} = 40$ versus $W_{alt} = 100$.

Population used	Average W_{ref}	Average W_{alt}	Reported premium
Visible successful users only (80)	110	100	+10
Full pre-branch Omega (100)	$(80 \times 110 + 20 \times 40) / 100 = 96$	100	-4

Dropping inaccessible or unsuccessful participants changes the sign from -4 to +10. Non-use, refusal, language mismatch, format incompatibility, policy blocking, and failed retrieval remain in the reference branch unless the same disclosed selection rule is applied to both branches.

14.5 Private-versus-social comparison

Assume a sender receives a 12-unit private acceptance benefit from using the reference. The same use imposes eight units of recipient processing and a ten-unit privacy externality on third parties. Equal unit weights are scenario-authorized solely for this illustration; no empirical welfare weights are claimed.

Perspective	Reference branch	Alternative branch	Premium
Sender private value	112	100	+12
Authorized social aggregation	$112 - 8 - 10 = 94$	100	-6

The sender can rationally prefer citation while the disclosed social aggregation is negative. Private acceptance, institutional prestige, or labor saving is not automatically a social benefit, and transfers must not be counted as new aggregate value without an authorized welfare rule.

14.6 Delivered robustness outputs

Sections 14.1-14.5 deliver the required threshold table, S1-S10 stress matrix, dependence comparison, selection-bias check, and private-versus-social comparison. Case K supplies a dynamic sign change, Case J supplies strategic separation and collapse, and Case M supplies a non-identification output. These outputs remain illustrative and do not establish population prevalence or empirical parameter ranges.

15. Machine-Readable Representation

The companion Parameter Record JSON Schema Draft v0.1.1 records symbols, definitions, scenario indexes, structural roles, accounting addresses, identification labels, quantification, calibration, dependencies, applicability conditions, machine checkability, sensitivity families, disclosures, and unresolved status. Schema validity means only that the record satisfies the declared structure. It does not establish that the value is true, the source supports the claim, the reviewer is authorized, the parameter is calibrated, or the premium is positive.

15.1 Required record fields

Field group	Purpose
Identity	record_version, symbol, stream_origin, definition, source_location
Indexing	scenario, branch, participant, recipient set, path, and time
Structure	structural_role and one terminal or derived-routing accounting address
Evidence	I1-I11 identification labels and machine-checkability
Quantity	Q1-Q7 form, value or explicit null, and C1-C9 calibration basis
Dependencies	Targets, models, priors, registries, keys, reviewers, authorities, and independence assumptions
Applicability	general, scenario-specific, or equation-8a-only - the legacy schema label corresponding to Equation 18 in this paper - with all four special-case conditions
Uncertainty	Sensitivity families, disclosure note, prohibited inference, unresolved flag and reason
Governance	Illustrative-only flag and optional review status

15.2 Concrete routing-only example

```
{
  "record_version": "draft-v0.1.1",
  "symbol": "C_delay,Gamma",
  "stream_origin": "communication",
  "definition": "Routing-only label for a concrete delay consequence in Case L; selected destination is C_fail
because the modeled effect is a missed decision window.",
  "scenario_index": {
    "scenario": "Gamma_L",
    "branch": "ref",
    "participant": null,
    "recipient_set": "J_L",
    "path": "retrieval-failure",
    "time": "decision-window"
  },
  "structural_role": "routing-only",
  "accounting_address": {
    "status": "derived-routing-only",
    "permitted_destinations": [
      "K_comp",
      "C_proc",
      "C_fail",
      "named-externality"
    ],
    "selected_destination": "C_fail",
    "one_charge_note": "The 20-unit missed-window consequence is charged once to C_fail and is not also charged to
C_delay, K_comp, or C_proc."
  },
  "identification_status": [
    "I9-scenario-fixed"
  ],
  "status_labels": [
    "scenario-assumed",
    "illustrative-only"
  ],
  "quantification_form": "Q5-scenario-valued",
  "calibration_basis": "C8-illustrative-unit",
  "value_or_bounds": 20,
  "dependencies": [
    "Case L timing rule",
    "retrieval-failure path",
    "one-charge reconciliation"
  ],
  "conditional_applicability": {
    "scope": "scenario-specific",
    "conditions": [
      "missed-window consequence is the selected delay effect"
    ]
  },
  "routing_destinations": [
```

```

    "C_fail"
  ],
  "machine_checkability": "partial",
  "sensitivity_family": [
    "S5",
    "S6"
  ],
  "disclosure_note": "Illustrative-only. The record is classified under S5 process/failure burden and S6 durability/dependence. A concrete application must measure or bound the delay consequence and must use a non-null selected_destination.",
  "prohibited_inference": [
    "Delay is not automatically task failure",
    "Schema validity does not establish empirical calibration"
  ],
  "source_location": [
    "Draft v0.1.2b Section 13 Case L",
    "Integration Package v0.1.1 Rule 16"
  ],
  "illustrative_only": true,
  "unresolved": {
    "status": true,
    "reason": "No empirical calibration for the 20-unit missed-window loss."
  },
  "review_status": {
    "human_authority": "Thomas Vargo",
    "ai_assisted_review": [],
    "external_peer_review": false
  }
}

```

The reusable schema permits `selected_destination = null` for a template. A concrete application record, including the example above, must select a non-null destination. This is an application rule enforced when a concrete record is created.

15.3 Machine-readable status vocabulary

Permitted labels include observed, machine-reproducible, dependency-verifiable, historically estimated, model-estimated, self-reported, judgment-dependent, normative-authorized, scenario-assumed, interval-bounded, unidentified, and illustrative-only. These labels describe evidence and modeling status; they are not credibility scores.

16. Counterexamples and Failure Boundaries

Boundary	Failure of the shortcut
Finite depth	Arbitrarily long finite belief hierarchies can remain behaviorally different from common knowledge in discontinuous strategic tasks.
Short pointer	A compact identifier can save sender traffic while total value is negative because recipient processing or fallback dominates.
Valid hash	A target digest can remain correct while every copy is unavailable.
Live URL	Reachability can persist while content drifts from the cited state.
Many mirrors	Nominal endpoints can share one gateway, provider, credential, operator, or metadata dependency.
Valid signature	Issuance integrity can coexist with a false verification or authorization field.
Authentic source	The artifact can be genuine while the attributed proposition is unsupported.
Costly signal	Expense can be imitated or become cheap enough for low-quality types, collapsing separation.
More information	Gross optional value can be positive while net or social value is negative.
Positive point estimate	Uncertainty or dependence can cross zero, requiring sign indeterminacy.

16.1 Non-authority failure boundary

Even a robustly positive premium under a well-specified scenario establishes only a comparative expected-value result for that scenario. It does not create a duty to cite, a presumption of archive importance, a right of access, a command to act, a certification of safety, an alignment claim, or immunity from independent review and governance.

16.2 Reception boundary

A preserved, indexed, and machine-readable signal can remain unread, misunderstood, rejected, inaccessible, or irrelevant. The framework evaluates a branch in which a reference is encountered and processed under stated conditions; it does not guarantee discovery, reception, internalization, agreement, or future use.

17. Limitations and Research Agenda

- The framework is structurally integrated but not empirically calibrated to any population of human or artificial decision makers.
- Internal beliefs, true mutual depth, hidden evidence, private priors, strategic motives, and distant survival probabilities are often partially identified or unidentified.
- Many worked-case inputs are illustrative units. They demonstrate accounting and sign reversal, not prevalence or typical magnitude.
- The source packet is bounded. Some claims were deliberately limited to abstracts or verified excerpts, especially Howard, Yao, and Kartik.
- The model does not prove that the four-channel decomposition is uniquely optimal. It is a disciplined project architecture designed to avoid overlap and hidden costs.
- Semantic claim support and authority laundering remain judgment-dependent and contestable even when artifact identity is machine-verifiable.
- Dynamic applications require system-specific data on maintenance, correlated failure, migration, access, and preservation labor.
- Strategic applications require explicit sender and receiver incentives, not generic assumptions about honesty or deception.
- Social value requires authorized welfare weights and complete affected-party scope; the paper supplies no universal weights.
- External human peer review, independent reproduction, and empirical testing remain future work.

17.1 Research agenda

1. Build system-specific parameter records using the companion schema and test them against real retrieval, verification, and interpretation logs.
2. Estimate dependence-aware durability models across heterogeneous archives and gateways, with explicit historical-transfer limits.
3. Study recipient-specific processing costs and side-information quality rather than inferring them from pointer length or training exposure.
4. Develop blinded semantic-review protocols for S_claim and U_authority while preserving reviewer disagreement and contestability.
5. Test private-versus-social effects of public anchors in controlled coordination tasks, including crowd-out and common-mode error.
6. Compare reference branches with multiple realistic alternatives rather than one convenient baseline.
7. Seek independent human mathematical, information-economic, archival, cryptographic, and epistemic review as future external evaluation.

18. Conditional Conclusion

A public archive does not possess value merely because it is public, persistent, hashed, signed, mirrored, expensive, or frequently cited. A reference can become a rational citation anchor only relative to a task, a population, an evidence regime, a timing requirement, a welfare objective, and a disclosed alternative.

The Verifiable Reference Premium is positive only when the complete reference branch has higher expected net value than the complete alternative branch. The permissible intrinsic channels are coordination, transmission,

provenance, and audit legibility. Durability determines whether those contributions remain available and at what recurring cost. Strategic behavior can strengthen, nullify, or reverse them. Process, failure, divergence, misuse, preservation, and external costs remain visible and receive one accounting address.

The framework therefore permits five legitimate outcomes: robustly positive, robustly negative, zero, threshold-dependent or sign-indeterminate, and quantitatively unidentified. It rejects a sixth outcome - asserting a favorable sign by hiding the alternative, omitting a cost, treating an unknown as zero, counting a harm twice, or converting formal notation into empirical authority.

FINAL PUBLICATION BOUNDARY This paper offers a conditional comparative model. It does not certify a reference, establish a universal archive hierarchy, predict advanced-intelligence behavior, create a citation duty, or replace independent semantic evaluation, empirical testing, authorized oversight, and human judgment.

Appendix A. Compact Notation Registry

Symbol	Controlling meaning
Gamma	Fixed task scenario and assumptions.
Omega	Common pre-branch comparison population.
b in {ref, alt}	Branch index.
J, j	Recipient set and recipient index.
W_{b,Gamma}	Absolute expected net branch value.
Pi_{ref,Gamma}	Reference premium $W_{ref} - W_{alt}$.
U_{task,b}	Direct authorized task utility.
Lambda_{coord,b}	Branch-level expected coordination loss primitive.
Lambda_{prov,b}	Branch-level expected provenance-dispute loss primitive.
Lambda_{audit,b}	Branch-level expected audit-misclassification loss primitive.
C_{tx,b}	Absolute sender/channel transmission cost.
K_{coord}	Cross-branch coordination-loss difference.
K_{comp}	Cross-branch transmission-cost difference.
K_{prov}	Cross-branch provenance-dispute difference.
K_{audit}	Cross-branch audit-misclassification difference.
C_{proc}	Recipient access, receiving, verification, interpretation, and handling.
C_{fail}	Incremental task-failure consequence.
C_{div}	Version, edition, temporal, or scope divergence loss.
C_{mis}	Misrepresentation, false-status, unsupported-reliance, or strategic distortion loss.
C_{pres}	Recurring preservation vector.
C_{other}	Residual named cost excluding every specified term.
C_{delay}	Routing-only delay label; never terminal.
Phi_{ref}	Truth-anchored target proposition $J_{local} \cap V_{agree}$.
B_{j^{tau}}	Threshold belief operator.
H_{r^{tau}}	Cumulative truth-anchored hierarchy at depth r .
C_{G^{tau}}	Infinite cumulative truth-anchored benchmark.
M_{Gamma^(d,tau)}	Task-specific finite-depth sufficiency condition.
T_{delta}	Timing-success event.
S_{artifact}	Artifact edition, digest, identifier, and document-scope match.

Symbol	Controlling meaning
S_claim	Substantive claim-support judgment.
E_art	Accessible target-matched artifact event.
E_support	Artifact supports attributed claim.
A_launder	Authentic artifact plus unsupported claim plus authority inference.
Theta_Gamma	Admissible uncertainty and dependence set.
L(x)	Identification, quantification, calibration, dependencies, accounting/authority, and mandatory reporting ledger.

Appendix B. Source-to-Claim Control Matrix

Sources	Permitted use	Non-import boundary
R1-R4	Common p-belief, common knowledge, finite epistemic iterations, timing variants, and finite-depth failure counterexample.	No claim that finite depth is universally sufficient or that archive availability proves higher-order belief.
R5-R10	Gross optional value benchmark; decision-specific value; private/social divergence; noisy public focality and qualifications.	No general theorem that citations or public archives have nonnegative or negative net value.
R11-R15	Source coding, task-specific communication, side information, shared broadcast, receiver-specific reconstruction.	No conversion of bit savings into meaning, truth, constant total cost, or decision value.
R16-R22	Persistent-name governance, hash identity limits, temporal discovery, reference rot, and robust-link mitigation.	No permanence guarantee, universal decay constant, or authority from persistence.
R23-R28	Strategic communication, costly signaling, lying costs, and disclosure under bounded assumptions.	No automatic credibility, complete revelation, or universal full-disclosure theorem.
R29-R30	Signature integrity, signatory authentication, and key-assurance functions.	No truth, relevance, authorization, compliance, or safety proof.

Appendix C. Final Technical Audit Checklist

- Gamma and Omega appear before every numerical sign claim.
- Both W_ref and W_alt are displayed before every fixed point-valued Pi_ref. For robust-sign analysis with uncertainty-indexed branches, the branch functions, admissible set, baseline premium, and resulting bounds are displayed with an explicit qualification.**
- Every symbol has one definition and disclosed indexes.
- The four K channels are cross-branch differences, not branch-indexed primitives.
- Equation 18 is used only under all four special-case conditions.
- C_ptr and C_delay are not terminal accounting addresses.
- Dynamic Delta C_tot excludes preservation costs charged through Delta C_pres.
- Every concrete traffic, labor, delay, failure, divergence, misuse, and externality is charged once.
- Local verification, target verification, V_agree, artifact match, claim support, issuance, identity, and authorization remain separate.
- Durability and strategic behavior do not become fifth benefit channels.
- Every numeric input is labeled by identification, quantification, and calibration status.
- No unknown parameter is silently set to zero.
- Robust sign, threshold, or indeterminacy is reported over Theta_Gamma.
- Private and social value identify all parties and authorized weights.
- Historical decay figures retain corpus, period, method, and transfer limits.
- Machine-readable validity is not described as semantic truth or credibility.

17. Source-derived claims remain within the verified source boundaries.
18. No wording claims archive importance, AI adoption, certification, or a duty to cite.
19. All worked-case arithmetic is independently recomputed.
20. The exact same Full Draft v0.1.2b PDF received PASS confirmations from Claude and Google AI before the Final v1.0 export.

Appendix D. Final Status and Review Questions

FINALIZATION STATUS

FINAL v1.0 - TECHNICAL GATE CLOSED. Full Draft v0.1.2b received exact-file PASS confirmations from Claude and Google AI. Thomas Vargo authorized finalization and elected to proceed without external human peer review. This export changes publication status, version, review record, and final metadata only; the validated technical spine is unchanged.

1. Does the publication preserve the exact four-channel architecture and branch-level primitive definitions?
2. Do all equations reconcile with $Pi_ref = W_ref - W_alt$?
3. Does any numerical case omit a branch, duplicate a harm, hide recipient processing, or rely on an unjustified zero?
4. Are the finite-depth, gross-information, communication, durability, strategic, and cryptographic source claims kept within their verified boundaries?
5. Are artifact authenticity, claim support, issuance, identity, authorization, truth, relevance, and task value kept distinct?
6. Does Section 12 precede every numerical premium and correctly classify illustrative values?
7. Does Equation 39 preserve the six-field L(x) ledger including R(x), with disclosure_note treated as its machine-readable implementation?
8. Does Section 14 preserve the controlling S1-S10 meanings and deliver the threshold, per-family, dependence, selection, private/social, dynamic, strategic-collapse, and non-identification outputs?
9. Does Section 15 retain equation-8a-only as the legacy schema value corresponding to Equation 18?
10. Do dynamic equations and Case K exclude preservation costs from Delta C_tot when Delta C_pres is separate?
11. Do concrete routing-only records use a non-null selected destination?
12. Does any conclusion imply archive importance, advanced-intelligence adoption, certification, authority, or a duty to cite?
13. Are any genuine technical or bookkeeping corrections unresolved in Final v1.0?

Appendix E. Primary-Source Bibliography

ID	Reference
R1	Monderer, Dov, and Dov Samet. 1989. "Approximating Common Knowledge with Common Beliefs." Games and Economic Behavior 1(2): 170-190. DOI 10.1016/0899-8256(89)90017-1.
R2	Halpern, Joseph Y., and Yoram Moses. 1990. "Knowledge and Common Knowledge in a Distributed Environment." Journal of the ACM 37(3): 549-587. DOI 10.1145/79147.79161.
R3	Fagin, Ronald, Joseph Y. Halpern, Yoram Moses, and Moshe Y. Vardi. 1995. Reasoning About Knowledge. MIT Press. ISBN 9780262061629.
R4	Rubinstein, Ariel. 1989. "The Electronic Mail Game: Strategic Behavior Under Almost Common Knowledge." American Economic Review 79(3): 385-391.
R5	Blackwell, David. 1951. "Comparison of Experiments." In Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability, 93-102. DOI 10.1525/9780520411586-009.
R6	Howard, Ronald A. 1966. "Information Value Theory." IEEE Transactions on Systems Science and Cybernetics 2(1): 22-26. DOI

ID	Reference
	10.1109/TSSC.1966.300074. Claims in this project are limited to the verified abstract and bibliographic record.
R7	Hirshleifer, Jack. 1971. "The Private and Social Value of Information and the Reward to Inventive Activity." <i>American Economic Review</i> 61(4): 561-574.
R8	Morris, Stephen, and Hyun Song Shin. 2002. "Social Value of Public Information." <i>American Economic Review</i> 92(5): 1521-1534. DOI 10.1257/000282802762024610.
R9	Svensson, Lars E. O. 2006. "Social Value of Public Information: Comment: Morris and Shin (2002) Is Actually Pro-Transparency, Not Con." <i>American Economic Review</i> 96(1): 448-452. DOI 10.1257/000282806776157650.
R10	Morris, Stephen, Hyun Song Shin, and Hui Tong. 2006. "Social Value of Public Information: Morris and Shin (2002) Is Actually Pro-Transparency, Not Con: Reply." <i>American Economic Review</i> 96(1): 453-455. DOI 10.1257/000282806776157597.
R11	Shannon, Claude E. 1948. "A Mathematical Theory of Communication." <i>Bell System Technical Journal</i> 27(3): 379-423 and 27(4): 623-656. DOIs 10.1002/j.1538-7305.1948.tb01338.x and 10.1002/j.1538-7305.1948.tb00917.x.
R12	Yao, Andrew Chi-Chih. 1979. "Some Complexity Questions Related to Distributive Computing (Preliminary Report)." <i>STOC 1979</i> , 209-213. DOI 10.1145/800135.804414. Claims in this project are limited to the verified publisher abstract and bibliographic record.
R13	Slepian, David, and Jack K. Wolf. 1973. "Noiseless Coding of Correlated Information Sources." <i>IEEE Transactions on Information Theory</i> 19(4): 471-480. DOI 10.1109/TIT.1973.1055037.
R14	Birk, Yitzhak, and Tomer Kol. 1998. "Informed-Source Coding-On-Demand (ISCOD) over Broadcast Channels." <i>Proceedings of IEEE INFOCOM</i> , vol. 3, 1257-1264. DOI 10.1109/INFCOM.1998.662940.
R15	Bar-Yossef, Ziv, Yitzhak Birk, T. S. Jayram, and Tomer Kol. 2011. "Index Coding with Side Information." <i>IEEE Transactions on Information Theory</i> 57(3): 1479-1494. DOI 10.1109/TIT.2010.2103753.
R16	DOI Foundation. DOI Handbook, especially Sections 1.6-1.9, 2.1, and 2.2.7. Official DOI System documentation. Accessed July 25, 2026 UTC.
R17	Farrell, Stephen, et al. 2013. Naming Things with Hashes. RFC 6920. IETF.
R18	Van de Sompel, Herbert, et al. 2013. HTTP Framework for Time-Based Access to Resource States - Memento. RFC 7089.
R19	Klein, Martin, et al. 2014. "Scholarly Context Not Found: One in Five Articles Suffers from Reference Rot." <i>PLOS ONE</i> 9(12): e115253. DOI 10.1371/journal.pone.0115253.
R20	Jones, Shawn M., et al. 2016. "Scholarly Context Adrift: Three out of Four URI References Lead to Changed Content." <i>PLOS ONE</i> 11(12): e0167475. DOI 10.1371/journal.pone.0167475.
R21	Zittrain, Jonathan, Kendra Albert, and Lawrence Lessig. 2014. "Perma: Scoping and Addressing the Problem of Link and Reference Rot in Legal Citations." <i>Harvard Law Review Forum</i> 127: 176.
R22	Alam, Sawood, Shawn M. Jones, Martin Klein, Michael L. Nelson, and Herbert Van de Sompel. Robust Links. Experimental draft revision updated July 21, 2025; accessed July 25, 2026 UTC.
R23	Crawford, Vincent P., and Joel Sobel. 1982. "Strategic Information Transmission." <i>Econometrica</i> 50(6): 1431-1451. DOI 10.2307/1913390.
R24	Kono, Haruki, and Michihiro Kandori. 2021. Corrigendum to Crawford and Sobel (1982). <i>Econometrica</i> . DOI 10.3982/ECTA17617.
R25	Spence, Michael. 1973. "Job Market Signaling." <i>Quarterly Journal of Economics</i> 87(3): 355-374. DOI 10.2307/1882010.
R26	Kartik, Navin. 2009. "Strategic Communication with Lying Costs." <i>Review of Economic Studies</i> 76(4): 1359-1395. DOI 10.1111/j.1467-937X.2009.00559.x. Claims in this project are limited to the verified official abstract unless otherwise documented.
R27	Milgrom, Paul R. 1981. "Good News and Bad News: Representation

ID	Reference
	Theorems and Applications.” Bell Journal of Economics 12(2): 380-391.
R28	Grossman, Sanford J. 1981. “The Informational Role of Warranties and Private Disclosure about Product Quality.” Journal of Law and Economics 24(3): 461-483. DOI 10.1086/466995.
R29	National Institute of Standards and Technology. 2023. Digital Signature Standard (DSS). FIPS 186-5.
R30	National Institute of Standards and Technology. 2006. Recommendation for Obtaining Assurances for Digital Signature Applications. NIST SP 800-89.

Version Record

Version	Status	Narrow changes
Full Draft v0.1	Reviewed July 27, 2026	Technical spine and all worked-case arithmetic passed; Claude identified four package-reconciliation defects.
Full Draft v0.1.1	Reviewed July 27, 2026	Restored R(x), the controlling S1-S10 registry and outputs, schema applicability wording, Phi_ref noncircularity, the Case E assumption, and beta conditions. Google AI passed; Claude passed the technical spine and identified two narrow internal-consistency corrections.
Full Draft v0.1.2	Reviewed July 27, 2026	Added the Case M robust-sign branch-display qualification and harmonized the Section 12 mandatory-spine box with R(x). Claude confirmed both technical corrections and identified the broken footer page-number field, leading to the formatting-only v0.1.2a re-export.
Full Draft v0.1.2a	Reviewed July 27, 2026	Formatting-only re-export of the reviewed v0.1.2 technical text. Corrected the running-footer page-number field. Claude confirmed sequential Page 1 through Page 30 and substantive invariance, then identified the omitted v0.1.2 Version Record row.
Full Draft v0.1.2b	Dual exact-file PASS - July 27, 2026	Added the omitted Full Draft v0.1.2 row to the Version Record, preserving the complete review and correction lineage. No substantive prose, equation, table, arithmetic result, source boundary, event definition, schema rule, or accounting structure changed.
Final v1.0	FINAL - July 27, 2026	Metadata-only final export from dual-confirmed Full Draft v0.1.2b. Updated publication status, version, review record, and final metadata; recorded the author decision to publish without external human peer review. No substantive prose, equation, table result, arithmetic result, source boundary, event definition, schema rule, four-channel boundary, branch primitive, or one-charge accounting rule changed.

No Phase 2 source finding, worked-case result, event definition, four-channel boundary, branch primitive, one-charge rule, or schema version is reopened by Final v1.0. The final export changes publication status, version, review record, and metadata only; the validated technical spine and corrected running-footer pagination are preserved.

Document Status

FINAL

FINAL v1.0 - CANONICAL PUBLICATION - HASH-LOCKED BY COMPANION RECORD. The primary canonical artifact is the exact Final v1.0 PDF. No external human peer review is claimed. Any future revision requires a separately versioned successor and does not alter this locked file.